

Junghoon Chung

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Summary

PhD candidate in Industrial and Systems Engineering specializing in **machine learning, causal inference, and multimodal human behavior modeling**. Experienced in developing **end-to-end pipelines and evaluation frameworks** that translate complex data into actionable, user-centered insights. Skilled in applying **AI/ML methods, statistical analysis, and experimental design** to real-world problems, with publications in human factors and ergonomics. Seeking a research or data science internship to develop adaptive, data-driven solutions.

Education

University of Washington, Seattle	Sep 2025 – Jun 2029 (Expected)
Ph.D in Industrial and Systems Engineering, Dean's Fellowship Recipient	
Virginia Tech, Blacksburg	Aug 2023 – May 2025
M.S in Industrial and Systems Engineering (Thesis Track), GPA 3.89/4.0	
Yonsei University, Seoul	Mar 2017 – Feb 2023
B.S in Industrial Engineering, GPA 3.52/4.0, Student Council President	

Experience

Graduate Research Assistant, Human Automation Systems Lab, University of Washington	Sep 2025 – Present
- Built ML models to assess vigilance using multimodal physiological time-series data (ECG, eye tracking, BVP, EDA, HR); developed preprocessing pipelines for heterogeneous sensor streams using Python and scikit-learn - Engineered 30+ physiological and contextual features across 10,000+ observations per participant; improved predictive performance by ~20% through feature engineering and model refinement	
Graduate Research Assistant, Occupational Ergonomics and Biomechanics Lab, Virginia Tech	Aug 2023 – May 2025
- Developed end-to-end data pipeline to collect, preprocess, and analyze multimodal datasets (EMG, EEG, IMU, motion capture, behavioral logs) across longitudinal studies (3 weeks/participant; 100,000+ records) - Designed and validated zero-shot NLP classifier for unstructured work-context text; built time-series forecasting models incorporating contextual variables; improved prediction accuracy through data segmentation and clustering - Identified behavioral outcome drivers and communicated findings through technical reports and data visualizations	

Key Projects

Human Vigilance Modeling with Causal Inference | University of Washington

Built predictive models of moment-to-moment alertness using multimodal physiological and contextual data (ECG, BVP, EDA, HR, eye tracking), **engineering 30+ features across 10,000+ observations** per participant. Improved model performance by **20% through feature engineering and hyperparameter tuning**. Applied causal inference frameworks to disentangle key drivers of performance variability and isolate actionable intervention targets.

Driver Response Modeling for V2X Pedestrian Alerts | City of Bellevue / University of Washington

Designed and conducted a field study collecting **physiological sensor data (E4 Empatica: EDA, BVP, HR, skin temperature), NASA-TLX workload ratings, and vehicle kinematics** to model driver cognitive and behavioral responses to crosswalk pedestrian alerts. Developed analysis pipeline linking sensor-derived physiological arousal to driving behavior outcomes in a connected vehicle safety context.

Context-Aware Sit-Stand Intervention | Virginia Tech

Analyzed longitudinal behavioral datasets (3 weeks/participant; 100,000+ records) to forecast posture and behavioral transitions during computer work. Developed **time-series forecasting models incorporating contextual variables** (task type, meeting schedule, environmental sensors) and improved prediction accuracy through **user clustering and behavior segmentation**. Published at HFES 2025; two journal manuscripts in preparation.

Skills

Programming: Python, SQL (MySQL), Swift (WatchOS)

ML/DL: Scikit-learn, TensorFlow, PyTorch

Data Analysis: Statistical modeling, Causal inference, Experimental design, Time-series analysis

Communication: Technical reporting, Cross-functional collaboration

Awards

Dean's Fellowship, College of Engineering	University of Washington, 2025–2029
Social-Innovation Capability Video Contest 2nd Place – Data mining	Yonsei University, 2022
Merit-based Scholarship (Academic Excellence)	Yonsei University, 2021